SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded

:

Analytical Problem Solving

(1) Compare TYPE-1 & TYPE 2-

Feature	Running :
TYPE-1	Hypervisor : Directly on Hardware
TYPE-2	Hypervisor on a host operating system
Performance	High comparitively lower
Security	: Smaller host OS attack surface Depends on host OS security
Management :	Suitable for large datacenters Easy for personal use and testing
Example :	VMware ESXI, Microsoft HYPER-V, VMare workstation pro, virtual box

Conclusion :

TYPE-1 hypervisors are better for mission Critical cloud Applications. because they Provide better Performance.

Workload Isolation in cloud Provider

Virtualization Allows finance, healthcare and student portal workloads to run on the same physical server using seperate virtual machines

- Isolation : Each VM has its own operating system memory space, & Allocated resources
- Attack surface 1 : Hypervisor : A vulnerability could Allow an Attacker to Escape one VM and affect others
- Attack surface 2 - virtual network : Incorrect network configuration could expose sensitive data between workloads

Conclusion :

Use strong VM Isolation, network, segmentation. Access control encryption and regular security updates.

scaling and automat

(3) Analyze VM Host utilization.

Metric	value	likely Problem
CPU Utilization	92.1.	CPU Overload
Memory Utilization	88 1.	Memory Pressure
Storage I/O latency	35 MS	Slow storage operations
VM boot time	140 s	Slow startup due to resource content- or storage delays.

Corrective actions:

Migrate some VMs to less-locate hosts.

Increases CPU and RAM capacity or optimize resource Allocation.

Upgrade Storage to faster SSD/NVMe or improve storage configuration.

Investigate startup service and storage bottlenecks to reduce boot time.

conclusion:

The host is overloaded and both CPU Pressure and high storage latency may be affecting VM performance

④ Software Defined Datacenter (SDDC)

An SDDC uses a software to manage and Automate datacenter resources instead of relying mainly on manual hardware configuration.

Compute virtualization.

Creates and manages VMs dynamically

Storage virtualization:

Combine storage resource into flexible storage pools.

network virtualization.

Creates software based network routing, and security policies

Advantages over a traditional datacenter

- Faster resource provisioning
- Easier scaling and automation
- Improved resource utilization
- Centralized management & policies enforcement

Conclusion:

SDDC improves agility by making compute, storage and networking programmable and easier to manage.

(5) Multi-cloud federation and identity mismatch.

Identity mismatches occur when cloud providers use different user identities, authentication methods, roles, permission. format

Proposed secure design:

Use a central Identity provider for Authentication

Implement Federation using SAML 2.0 or open ID connect (OIDC)

Map user identities and roles consistently Accross Providers.

Apply role-based Access control and least privellage permission

Use MFA, encrypted communication, & short lived Access tokens

conclusion:

Identity Federation Providers. Consistent Authentication across clouds while reducing duplicate Accounts and protecting sensitive information.